

Table of contents

The Essentials	1
Why Study High-Dimensional Spaces?	1
The Representation Space	2
The Curse of Dimensionality	2
1. Data Sparsity	2
2. Empty Sphere and Spiky Cube	3
3. Volume Distribution	4
4. The Gaussian Egg	5
5. Distance Concentration	5
6. Quasi-Orthogonality	6
7. Hubness	7
8. Concentration Inequalities	8
The Blessing of Dimensionality	9
1. Laws of Large Numbers	9
2. Random Projections	10
Impact in Data Science	11
1. Sampling	11
2. Indexing	11
3. Learning	12
4. Why Does Data Science Still Work?	12
Global Summary	13
Positive Aspects (Blessing)	13
Negative Aspects (Curse)	13
Unique Solution	14

The Essentials

Why Study High-Dimensional Spaces?

Data Science studies how to represent the real world in the form of numerical data within mathematical spaces. It focuses on the **geometric** and **statistical** properties of these spaces.

Universal Pipeline:

- Real world $\rightarrow$ Sensors $\rightarrow$ Raw data $\rightarrow$ Numerical representation
- Everything becomes a **matrix**: N samples, D features

Two Complementary Perspectives:

- **Statistical**: Sample from an underlying distribution

- **Algebraic:** Matrix $X \in \mathbb{R}^{D \times N}$
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The Representation Space

A D -dimensional space has a complete structure:

- Inner product $\rightarrow$ Norm $\rightarrow$ Distance $\rightarrow$ Metric space

Central Question: What happens when D becomes large?

- Negative effects: **Curse** of dimensionality
 - Positive effects: **Blessing** of dimensionality
-

The Curse of Dimensionality

1. Data Sparsity

To accurately estimate a distribution in D dimensions, an **exponential** number of samples is required:

$$N \approx n \cdot b^D$$

where n = samples per bin, b = number of bins per dimension.

Example:

- $D = 6$, $b = 10$, $n = 10 \rightarrow N \approx 10^7$ samples needed
- With fixed N : most bins remain **empty**

Details: Estimation Quality

If N is fixed, the estimation quality becomes:

$$n = \frac{N}{b^D} \implies \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b^D}$$

The probability that a bin is occupied decreases exponentially with D .

Quick Review: Data Sparsity

Requires $N \approx n \cdot b^D$ samples (**exponential** growth).

If N is fixed: $n = N/b^D \rightarrow$ most bins **empty**.

2. Empty Sphere and Spiky Cube

In a high-dimensional cube, the volume of the inscribed sphere becomes **negligible**:

$$\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1} \Gamma(D/2)} \xrightarrow{D \rightarrow \infty} 0$$

Consequences:

- Volume concentrates in the **corners** of the cube
- Center-corner distance = $\frac{\sqrt{D}}{2}$ (grows with $\sqrt{D}$)
- Uniform sampling $\rightarrow$ **empty** sphere

Details: Volume of the Hypersphere

General Formula:

$$\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D \Gamma(D/2)}$$

Recurrence Relation (for $D > 1$):

$$\text{vol}(B_D(r)) = \frac{2\pi r^2}{D} \text{vol}(B_{D-2}(r))$$

Corners of the Unit Cube:

- Number of corners: 2^D
- Distance to origin: $\|x_i\|_2 = \frac{\sqrt{D}}{2}$

D	1	2	4	5	100
$\frac{\sqrt{D}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{5}}{2}$	5

Quick Review: Empty Sphere & Spiky Cube

Sphere/cube ratio $\rightarrow 0$. Volume concentrated at **corners** (distance $\sqrt{D}/2$).
Uniform sampling $\rightarrow$ **empty** sphere.

3. Volume Distribution

For **any shape** (cube, sphere, or other), volume concentrates near the surface:

$$\lim_{D \rightarrow \infty} \frac{\text{vol}(V_D(r)) - \text{vol}(V_D((1-\varepsilon)r))}{\text{vol}(V_D(r))} = 1$$

Principle:

- Reducing size by factor $(1-\varepsilon) \rightarrow$ volume reduced by $(1-\varepsilon)^D \rightarrow 0$
- Almost all volume is in a thin **shell** at the surface

Details: Calculation for Hypercube and Hypersphere

For the Hypercube:

$$\frac{\text{vol}(C_D((1-\varepsilon)r))}{\text{vol}(C_D(r))} = (1-\varepsilon)^D \xrightarrow{D \rightarrow \infty} 0$$

For the Hypersphere:

$$\frac{\text{vol}(B_D((1-\varepsilon)r))}{\text{vol}(B_D(r))} = (1-\varepsilon)^D \xrightarrow{D \rightarrow \infty} 0$$

Technical Note: $(1-\varepsilon)^D \leq e^{-D\varepsilon} \rightarrow 0$ as $D \rightarrow \infty$

Decomposition Method: Divide the volume into arbitrarily small hypercubic voxels, then apply the same reasoning.

Quick Review: Volume Distribution

For **any shape:** ratio $(1-\varepsilon)^D \rightarrow 0$.
Volume **concentrated at the surface**.

4. The Gaussian Egg

For a distribution $X \sim \mathcal{N}(0, \text{Id}_D)$, the radius $R = \|X\|_2$ follows:

$$R^2 = \sum_{i=1}^D X_i^2 \sim \chi^2(D) \implies \mathbb{E}[R^2] = D \implies \mathbb{E}[R] \approx \sqrt{D}$$

The density is **concentrated** around the radius $\sqrt{D}$.

Gaussian Annulus Theorem:

$$P\left[\left|\|X\|_2 - \sqrt{D}\right| > \beta\right] \leq 3e^{-c\beta^2}$$

Almost all density is in a **thin ring** around $\sqrt{D}$.

Details: Density Along the Radius

Context: Coordinates $X_i \sim \mathcal{N}(0, 1)$ i.i.d.

Integration along r = take the norm. R is the random variable of the radius.

Alternative Formulation:

$$P\left[\left|\frac{\|X\|}{\sqrt{D}} - 1\right| \leq \varepsilon\right] \approx 1$$

For a Gaussian $\mathcal{N}(0, \sigma^2 \text{Id}_D)$, the density is concentrated at radius $\sigma\sqrt{D}$.

Quick Review: Gaussian Egg

For $X \sim \mathcal{N}(0, \text{Id}_D)$: $R^2 \sim \chi^2(D)$ so $\mathbb{E}[R^2] = D$.

Density concentrated at radius $\sqrt{D}$ (**thin ring**).

5. Distance Concentration

All neighbors become **equidistant** as D increases.

Theorem: If the variance of the normalized distance tends to 0, then:

$$\lim_{D \rightarrow \infty} P\left[\frac{D_{\max} - D_{\min}}{D_{\min}} \leq \varepsilon\right] = 1$$

where $D_{\min}$ and $D_{\max}$ are the distances to the nearest and farthest neighbors.

Consequences:

- All neighbors appear at distance $D_{\min}$
- Discriminative power decreases exponentially
- KNN and ε -NN become **non-significant**

Details: Concentration Condition

Complete Theorem: If $\lim_{D \rightarrow \infty} \text{Var} \left[\frac{d(x,y)^p}{\mathbb{E}[d(x,y)^p]} \right] = 0$ for $0 < p < \infty$, then concentration occurs.

This condition means that the relative fluctuations of the distance become negligible.

Quick Review: Distance Concentration

If normalized distance variance $\rightarrow 0$, then $\frac{D_{\max} - D_{\min}}{D_{\min}} \rightarrow 0$.
All neighbors **equidistant**. KNN loses meaning.

6. Quasi-Orthogonality

In high dimensions, random vectors exhibit surprising geometric properties:

Two random vectors are quasi-orthogonal:

$$\mathbb{E}[\cos(\theta)] = 0 \implies \theta \approx \frac{\pi}{2}$$

Random triangle is quasi-equilateral:

$$\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \approx \frac{\pi}{3}$$

Consequence: On a sphere, volume concentrates at the **equator** (regardless of direction).

Details: Angle Calculation Between Vectors

For two random vectors:

Define 4 i.i.d. random variables W, X, Y, Z on $\Omega \subset \mathbb{R}^D$:

$$\mathbb{E}[\cos(\angle(X - Y, Z - W))] = \mathbb{E} \left[\frac{\langle X - Y, Z - W \rangle}{\|X - Y\| \cdot \|Z - W\|} \right]$$

Since W, X, Y, Z are i.i.d., $X - Y$ and $Z - W$ are also i.i.d. and centered: $\mathbb{E}(X - Y) = \mathbb{E}(Z - W) = 0$.

The numerator is $\text{cov}(X - Y, Z - W) = 0$.

For a triangle:

With 3 i.i.d. random variables X, Y, Z , compute:

$$\frac{\langle X - Y, Z - Y \rangle}{\langle X - Y, X - Y \rangle} = \frac{\langle X, Z \rangle - \langle X, Y \rangle - \langle Y, Z \rangle + \langle Y, Y \rangle}{\langle X, X \rangle + \langle Y, Y \rangle - 2\langle X, Y \rangle}$$

In expectation: $= \frac{1}{2}$

Note: To sample on the unit hypersphere, sample a centrally symmetric distribution (e.g., Gaussian) and normalize.

Quick Review: Quasi-Orthogonality

2 random vectors: $\theta \approx \pi/2$ (quasi-orthogonal).

Random triangle: $\theta \approx \pi/3$ (equilateral).

Sphere: volume at the equator.

7. Hubness

Some samples appear in the neighborhood of **many** others.

Definition: $n_k(x_j)$ = number of times x_j is a k-NN of another point.

Observation: The distribution of n_k is **asymmetric**:

- A few points (**hubs**) have very high values
- Most points have low values
- These hubs are “close” to all other samples

Details: Formal Definition of Hubness

Given X , define:

$$r_{ik}(x_j) = \begin{cases} 1 & \text{if } x_j \in V_X^k(x_i) \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$n_k(x_j) = \sum_i r_{ik}(x_j)$$

Empirical Visualization: For 20-NN with $D = 100$ and 1000 uniform samples on 50 bins, a strong asymmetry in the distribution of n_k is observed.

Quick Review: Hubness

$n_k(x_j)$ = number of times x_j is a k-NN of another point.

Asymmetric distribution: few **hubs** in many neighborhoods.

8. Concentration Inequalities

Concentration inequalities mathematically explain these phenomena.

Fundamental Inequalities:

- **Markov:** $P(X > \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon}$
- **Chebyshev:** $P(|X - \mu| > \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$

Hoeffding's Inequality: For $\{X_i\}_{i \in [D]}$ independent with $P(a \leq X_i \leq b) = 1$ and $\mathbb{E}[X_i] = \mu$:

$$P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$$

where $\bar{X} = \frac{1}{D} \sum_{i=1}^D X_i$

Explanation:

- Aggregation of independent variables $\rightarrow$ convergence to μ
- Minkowski distances have structure $\sum_{i=1}^D (\cdot)$
- For $d_p(x, y)$ to be small (large), **all** D coordinates must be similar (dissimilar)
- As D increases, this becomes unlikely

Details: Application to Distances

Minkowski distances:

$$d_p(x, y) = \left(\sum_{i=1}^D |x[i] - y[i]|^p \right)^{1/p}$$

Hoeffding's inequality formally explains why distance values **concentrate**.

Intuition: For large D , extreme values of $d_p(x, y)$ become unlikely because they require coordination across **all** dimensions.

Note: Example of PAC (Probably Approximately Correct) analysis.

Quick Review: Concentration Inequalities

Hoeffding: $P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$.
Structure $\sum_{i=1}^D (\cdot)$ in distances $\rightarrow$ **concentration**.

The Blessing of Dimensionality

1. Laws of Large Numbers

Weak Law of Large Numbers (WLLN):

For $\{X_i\}_{i \in [D]}$ i.i.d. with $\mathbb{E}[X_i] = \mu$:

$$\lim_{D \rightarrow \infty} P(|\bar{X} - \mu| < \varepsilon) = 1$$

where $\bar{X} = \frac{1}{D} \sum_{i=1}^D X_i$

Interpretation:

- The empirical mean $\bar{X}$ converges to the true expectation μ
- Fundamental basis of estimation theory
- Consistent with the frequentist approach to probabilities

Central Limit Theorem (CLT):

$$Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$$

if $\text{Var}(X_i) = \sigma^2$

Details: Proof via Chebyshev

The WLLN can be proven from Chebyshev's inequality:

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X})}{\varepsilon^2} = \frac{\sigma^2}{D\varepsilon^2} \xrightarrow{D \rightarrow \infty} 0$$

The CLT gives the form (Gaussian) and the rate ($1/\sqrt{D}$) of convergence.

Quick Review: Laws of Large Numbers

WLLN: $\bar{X} \xrightarrow{P} \mu$ as $D \rightarrow \infty$.

Basis for density estimation.
CLT: convergence rate $1/\sqrt{D}$.

2. Random Projections

Principle: Exponentially many quasi-orthogonal directions in high dimensions.

Johnson-Lindenstrauss Lemma:

Given $X \subset \Omega$, there exists a linear projection $p : \mathbb{R}^D \rightarrow \mathbb{R}^d$ such that:

$$(1 - \varepsilon)\|x_i - x_j\| \leq \|p(x_i) - p(x_j)\| \leq (1 + \varepsilon)\|x_i - x_j\|$$

with

$$d = O\left(\frac{\log N}{\varepsilon^2}\right)$$

Important Note: d does **NOT** depend on D !

Applications:

- Locally Sensitive Hashing (LSH) for approximate nearest neighbor (ANN) search
- Random bases for signal decomposition and coding

Caveat: Convergence rate slower than most concentrations

Details: Why It Works

Recall: Two random vectors u and v are likely orthogonal: $\langle u, v \rangle = \varepsilon$ (small).

Key Property: There are exponentially many quasi-orthogonal directions in high-dimensional spaces.

This allows creating random bases (quasi-orthonormal) that preserve essential metric properties while drastically reducing dimension.

Quick Review: Random Projections

Johnson-Lindenstrauss: projection $\mathbb{R}^D \rightarrow \mathbb{R}^d$ preserving distances with $d = O(\log N / \varepsilon^2)$ (independent of D).

Application: LSH for ANN.

Impact in Data Science

1. Sampling

Consequences of Sparsity:

- Requires an **exponential** number of samples to cover the space
- Concentration inequalities are **precision indicators** for estimation
- Rejection rate in rejection sampling → close to 100%

Details: Rejection Sampling

In practical processes like rejection sampling, the rejection rate becomes exponentially close to 100% due to the empty sphere phenomenon.

If sampling in a particular region (like a sphere inscribed in a cube), the acceptance probability decreases exponentially with D .

2. Indexing

Exploiting Structure:

- Indexing exploits data structure to anticipate which parts of the space to visit
- If all neighbors are at equal distance, the structure **disappears**

Spatial Partitioning Techniques (e.g., kd-trees):

- Attempt to uniquely identify each data point in a cell
- With distance concentration: cell size → full radius
- Search becomes **exhaustive** → complexity $O(N)$
- Hubness makes some data points answers to almost all queries

Details: Search Trees

kd-trees attempt to partition the space so that each data point is uniquely identified within a cell intersection (paths in the tree).

With distance concentration:

- The cell must be as large as the sphere's radius to cover the entire space
 - Pruning conditions become ineffective
 - The tree degenerates into a linear list
-

3. Learning

Machine Learning = discovering function approximators.

Problem: Find $\phi_\theta \in \mathcal{F}$ approximating $\phi : X \rightarrow Y$ by minimizing an error.

Catastrophic Theoretical Bound:

If $\mathcal{F}$ is a class of c -uniformly Lipschitz functions on Ω :

$$\sup_{\phi_\theta \in \mathcal{F}} \|\phi_\theta - \phi\|_\infty \geq \frac{c\sqrt{D}N^{-1/D}}{2} \sqrt{\frac{2}{\pi e}} \left(1 + O\left(\frac{\log D}{D}\right)\right)$$

For an error $c\varepsilon$ with $\varepsilon > 0$, the required number of examples is:

$$N \geq \frac{\varepsilon^{-D} D^{D/2}}{(2\pi e)^{D/2}}$$

Concrete Examples:

- $D = 3, \varepsilon = 0.1 \rightarrow N \geq 74$
- $D = 10, \varepsilon = 0.1 \rightarrow N \geq 687 \times 10^6$
- $D = 15, \varepsilon = 0.1 \rightarrow N \geq 377 \times 10^{12}$
- $D = 10, \varepsilon = 0.01 \rightarrow N \geq 6.9 \times 10^{18}$

Realistic Case: $D \sim 100, \varepsilon \sim 10^{-4} \rightarrow N \sim 10^{400}$ **IMPOSSIBLE**

Quick Review: Importance in Data Science

Sampling: Exponential need, rejection rate $\rightarrow 100\%$.

Indexing: Structure disappears, kd-trees $\rightarrow O(N)$, hubness.

Learning: $N \geq \varepsilon^{-D} D^{D/2} / (2\pi e)^{D/2} \rightarrow$ impossible for large D .

4. Why Does Data Science Still Work?

Assumptions in the Above Analyses:

- **Uniform** (or isotropic) data distribution
- **Independent** features

These assumptions are FALSE in practice!

Reality:

- Real data **do NOT** populate the space uniformly
- Features show **correlations**
- Data occupy **subspaces** of lower intrinsic dimensionality

Practical Solutions:

- Decorrelate features (linear latent space)
- Discover subspaces (clustering, nonlinear dimensionality reduction)
- PCA, autoencoders, neural networks
- Exploit underlying geometric structure

Key Insight: If features are informative, they are not independent, and data concentrate on manifolds of much lower dimension than D .

Quick Review: Why It Works

Analyses assume **uniform** distribution and **independent** features.

Reality: Data lie on **subspaces** of low intrinsic dimensionality.

Solutions: Decorrelation, clustering, dimensionality reduction.

Global Summary

High-dimensional geometry ($D \gtrsim 10$) is **different** from familiar 3D geometry.

Positive Aspects (Blessing)

- **Concentration inequalities:** Basis of statistical estimation
- **Laws of Large Numbers:** Enable density estimation
- **Random projections:** Exploit quasi-orthogonality

Negative Aspects (Curse)

- Data highly **sparse** (exponential sample requirement)
- **Concentration of distances and angles:**
 - All neighbors equidistant
 - Everything appears orthogonal
- Density concentrated on **narrow radii** $\rightarrow$ biased sampling
- **Hubs:** Part of all classes/responses

Unique Solution

Since this is by construction, the only solution is to operate in **low-dimensional spaces**:

- Dimensionality reduction
- Subspace discovery
- Feature decorrelation

Key Message: Data Science works because real data **are not uniform** and live on **low-dimensional submanifolds**.

Quick Review: Summary

High dimension ($D \gtrsim 10$) 3D geometry.

: Concentration inequalities, WLLN, random projections.

: Data sparsity, distance/angle concentration, hubs.

Solution: Operate in **low dimension** (PCA, autoencoders, structure exploitation).